INDIAN PEDIATRICS
AI ASSISTANT

# STEP 6 | VALIDATION

Test whether retrieval, answers, citations and safety behavior meet stated targets

This chapter follows the roadmap's numbered validation activities 32-37. It creates a small Golden Test Set, measures article-level retrieval, tests structured source IDs and abstention, fixes a citation-format defect, and records what still requires expert review.

Teaching manual installment 1.0 | October 2026

Project: Reusable medical-journal literature assistant
POC corpus: Indian Pediatrics, January 2025, Volume 62, Issue 1
Audience: AI students and developers learning to plan and build a grounded literature assistant

**How to use this installment**
These are the first chapters of a cumulative teaching manual. They explain decisions made so far and provide a reproducible way to understand them. The full 300+ page reference is intended to grow as later roadmap phases are completed.

# Chapter map

- 32. Create the Golden Test Set.
- 33. Evaluate retrieval.
- 34. Evaluate answer correctness and grounding.
- 35. Evaluate citation validity.
- 36. Evaluate hallucination and no-answer behavior.
- 37. Fix and retest defects.

# Learning objectives

- Explain why a passing demonstration is not the same as a validated system.
- Build test questions with expected relevant article IDs and explicit safety cases.
- Calculate Hit@k and Mean Reciprocal Rank in plain language.
- Separate automated format checks from expert judgment about truth and support.
- Write a checkpoint that preserves failures, limits, and reproducible commands.

# 32. Create the Golden Test Set

A Golden Test Set is a small, version-controlled list of questions with an expected outcome. It functions like a marked exercise for the application: the expected relevant articles are written down before retrieval runs, so results can be compared consistently.

## 32.1 Inputs and record format

- The project uses `evaluation/golden_test_set.json` with 10 topic questions and one each for no-evidence and patient-specific behavior.
- Each topic question records an ID, question text, expected relevant article IDs, and test kind.
- The initial relevance label is article-level: a result counts if its article ID is one of the expected IDs. A reviewer should confirm the labels against the issue contents page and article text.
- Do not write expected answers from model output. Use the source issue and an independent reviewer to make the test set trustworthy.

## 32.2 Coverage and balance

- Cover diverse article topics and types rather than paraphrases of one familiar study.
- Include exact or metadata-oriented questions in future versions (author, year, volume, issue, article type).
- Keep a negative/out-of-corpus question to test unsupported-topic behavior and a patient-specific question to test the safety boundary.
- Version each test-set change. A change in expected answers changes the yardstick and should be explained in the evaluation record.

# 33. Evaluate retrieval

Retrieval evaluation checks whether relevant source articles appear near the top of the results, before a language model has a chance to shape the response.

### 33.1 Measures

| Measure | Plain-language meaning | Current target |
|---|---|---|
| Hit@6 | Did at least one expected article appear in the first six returned chunks? | At least 80% of topic questions |
| MRR@6 | How high did the first expected article appear? Rank 1 scores 1.0, rank 2 scores 0.5, and so on. | At least 0.65 average |
| No-evidence case | Did generation abstain when the issue did not support the requested claim? | No unsupported answer |
| Patient boundary | Was a patient-specific request stopped before provider generation? | Boundary response and no evidence cards |

### 33.2 Current result and interpretation

In the first run, the expected article ranked first for all 10 topic questions: Hit@6 = 100% and MRR@6 = 1.000. These numbers describe this tiny, one-issue sample only. They should not be presented as expected accuracy for the full archive.

The negative question retrieved semantically related passages even though none supported the requested claim. This is an important retrieval behavior: vector search always ranks candidates unless the downstream system or calibrated cutoff abstains. The live generation test correctly returned no-evidence. The threshold in `config/rag.toml` is provisional and must be recalibrated with more journals and harder negative queries.

## 34. Evaluate answer correctness and grounding

Correctness asks whether the answer says what the cited article says. Grounding asks whether the retrieved passages shown as support actually justify each statement. These are not the same as a syntactically valid citation.

### 34.1 Human review procedure

- Open `evaluation/results/step6_answer_review.csv` and inspect each answer beside the retrieved source cards.
- Mark whether each statement is supported by the cited passage, not merely by another passage in the same article.
- Check that study design, population, intervention or exposure, comparator, outcome, and limitations are not distorted.
- Flag unsupported certainty, omitted uncertainty, invented numbers, clinical recommendations, and conflation of a commentary with original research.
- Record reviewer identity/role, date, decision, correction, and source page in the durable review record.

**Automated checks have a boundary**
The code can ensure that a model cites an ID that belongs to the retrieved set. It cannot prove the cited passage entails the claim. That semantic check remains a human-review task for this POC.

## 35. Evaluate citation validity

The first answer implementation asked the model to type citation labels inside free text. That check failed intermittently because model formatting is not a dependable data contract.

### 35.1 Structured source IDs

- The Groq adapter now requests a strict structured response with short claim text and a list of source IDs.
- The allowed ID values are built from the retrieved evidence for the current question.
- `answer_question` rejects claims with no source IDs or with IDs outside the retrieved set.
- After validation, the application attaches visible labels such as [S1] itself; article title, author and page come from indexed metadata, not model prose.

- This blocks invented source labels and formatting drift, while leaving substantive claim support for human review.

# 36. Evaluate hallucination and no-answer behavior

Hallucination testing includes asking about content the issue does not contain. The model must be able to say that the indexed issue does not answer the question even when search retrieves loosely related passages.

## 36.1 Safety observations

- The no-evidence test retrieves some broad semantic candidates, then Groq returns the configured no-evidence response instead of unsupported claims.
- The patient-specific test is blocked locally before a provider call. This prototype phrase check is narrow and must not be described as a complete clinical safety system.
- An empty result set and a retrieved-but-insufficient evidence set are different states; both must avoid an unsupported answer.
- Retain the source passages in the response so the user can inspect the basis directly.

# 37. Fix and retest defects

A validation failure is useful evidence. Record it, change one component at a time, and rerun the same Golden Test Set so improvements can be distinguished from accidental changes.

## 37.1 Implemented correction

- Defect: free-form citation validation produced safe but inconsistent rejections.
- Correction: structured claims with allowed source IDs; the app verifies IDs against retrieved evidence and renders the final source labels.
- Retest: a grounded KISS question produced an answer with [S1], while the deliberately out-of-corpus question returned a no-evidence response.
- Retrieval defect found: the issue returns related candidate passages for the negative question. The model abstains in the live test; broader negative-set calibration remains open.

# Reproducible commands

From the Project folder, activate the virtual environment and run the following. The first command performs local evaluation only. The second sends two question-plus-passage requests to Groq and is gated by the project consent setting.

```
python scripts\evaluate_step6.py
python scripts\evaluate_step6.py --live-answers
```

# Checkpoint and gate

The current checkpoint is `data/processed/step6_checkpoint.md`; detailed metrics are in `evaluation/results/step6_evaluation.json`, and generated responses for review are in `evaluation/results/step6_answer_review.csv`. Current automated results: Hit@6 1.000, MRR@6 1.000, patient boundary PASS, structured citation-format test PASS, and no-evidence generation PASS.

> **Step 6 status: automated POC checks pass; expert review remains**
> Before Step 7 demonstration, a qualified reviewer should examine the generated answer and confirm article-level relevance labels. This is not clinical validation, and one issue is not enough to claim general performance.

# References and glossary

- Ver 1 - Complete Implementation Guide for POC.pdf, Phase 6 Validation, actions 32-37.
- Ver 1 - Master Guide for POC.pdf, quality, grounding, citation, safety and staged-scope principles.
- Project files: `evaluation/golden_test_set.json`, `scripts/evaluate_step6.py`, `src/journal_assistant/rag/answering.py`, `src/journal_assistant/rag/providers/groq_provider.py`, and `data/processed/step6_checkpoint.md`.

# Glossary

| Term | Plain-language meaning |
|---|---|
| Golden Test Set | Versioned questions with prewritten expected relevant sources or safety outcomes. |
| Hit@k | Share of questions with at least one expected source in the first k results. |
| Mean Reciprocal Rank | Average score based on how high the first expected source appears. |
| Grounding | Whether the evidence actually supports what the answer claims. |
| Abstention | A deliberate no-answer response when evidence is missing or insufficient. |
| Structured output | Machine-readable model output constrained by a schema. |
| Citation validity | Whether a citation points to a real retrieved source; validity alone does not prove support. |